

LAB # 5 : Binomial and Poisson Distribution

Introduction

This lab is concerned with two important discrete probability distributions namely Binomial and Poisson. These discrete probability distribution can be represented graphically by a barplot or histogram. You may find it convenient to print the pdf version of this lab rather. This lab will take one session.

Binomial Distribution

Bernoulli trial can result in a success with probability p and a failure with probability $q = 1 - p$. Then the probability distribution of the binomial random variable X , the number of successes in n independent trials, is

$$f(x) = P(X = x) = \binom{n}{x} p^x q^{n-x}, \quad x = 0, 1, \dots, n.$$

The mean of the distribution is np and variance is npq . There are four basic functions d, p, q, r in R for calculating the probability in binomial distribution.

- $dbinom(x, n, p)$ returns $P(X = x)$ for binomial distribution. If x is a vector of integers, then R compute the probability for each value in x .

```
dbinom(0:3, 5, 0.5)
[1] 0.03125 0.15625 0.31250 0.31250
args("dbinom")
function (x, size, prob, log = FALSE)
NULL
```

- $pbinom(x, n, p)$ returns the cumulative probability $P(X \leq x)$ for binomial distribution.

```
pbinom(3, 5, 0.5)
[1] 0.8125
pbinom(0:5, 5, 0.5)
[1] 0.03125 0.18750 0.50000 0.81250 0.96875 1.00000
```

- $qbinom(p, n, p)$ returns the smallest value x such that $pbinom(x, n, p) \geq p$. That is, it computes quantiles for the binomial distribution.

```
qbinom(0.25, 10, 0.5)
[1] 4
qbinom(c(0.25, 0.5, 0.75), 10, 0.5)
[1] 4 5 6
pbinom(3:7, 10, 0.5)
[1] 0.1718750 0.3769531 0.6230469 0.8281250 0.9453125
```

- $rbinom(N, n, p)$ generates N random values from binomial distribution.

```

rbinom(10,4,0.5)
[1] 1 2 0 2 1 3 3 1 3 2

```

Example 1: Let us consider few examples.

- (a) According to recent survey, 65% of Americans prefer dogs to cats. If 8 American are chosen at random, what is the probability that 6 prefer dogs? That fewer than 6 do?
- (b) Let X represent the number of sixes in 10 rolls of a fair die. Simulate 50 runs of this probability experiment.
- (c) A weighted coin has a 42% chances of coming up heads. What is the expected number of heads in 5 tosses? The standard deviation? Construct a probability histogram for X , the number of heads in 5 tosses.

Sol.

```

(a)      dbinom(6,8,0.72)
[1] 0.3058222
pbinom(5,8,0.72)
[1] 0.3972716
(b)      rbinom(50,10,1/6)
[1]  4 1 1 1 2 1 1 1 2 0 1 1 2 2 0 0 0 2 3 0 1 1 3 1 4 0 2 4 1 2 2 1 2 2
[35] 0 1 1 1 5 4 1 0 1 3 4 1 2 1 1 1
table(rbinom(50,10,1/6))
  0  1  2  3  4
  5 19 13  8  5
qbinom(c(0.25,0.5,0.75),10,0.5)
[1] 4  5  6
pbinom(3:7,10,0.5)
[1] 0.1718750 0.3769531 0.6230469 0.8281250 0.9453125
(c)      x<-0:5
[1] 0 1 2 3 4 5
p<-dbinom(x,5,0.42)
[1] 0.06563568 0.23764642 0.34417757 0.24923203 0.09023918 0.01306912
sum(p)
[1] 1
weighted.mean(x,p)
[1] 2.1
weighted.mean((x-2.1)^2,p)
[1] 1.218
weighted.mean((x-2.1)^2,p)^0.5
[1] 1.10363
hist(p)

```

Example 2: Take $n = 10$ trials and compute the probability mass function for each $x = 0 : 10$. Sketch a graph (barplot) for the outcomes. Also make a graph for the cumulative probability.

Sol.

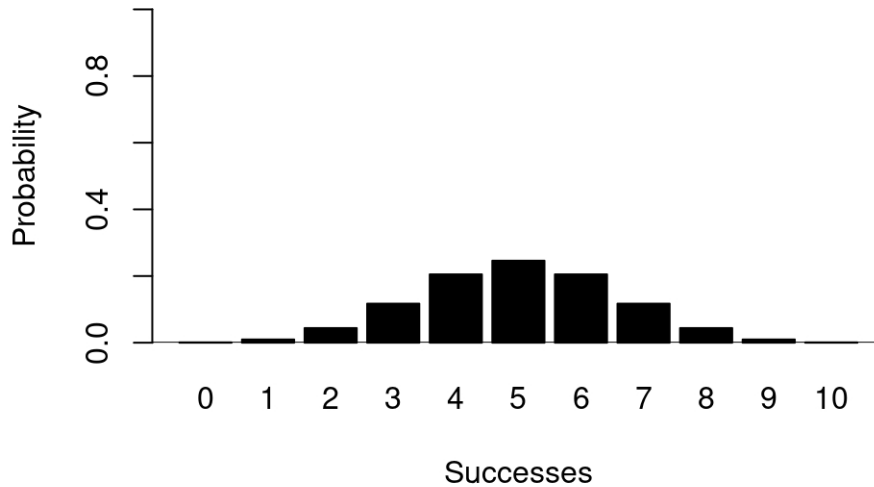
```

# graph of binomial probaility
binom.prob<-dbinom(0:10,10,0.5)
barplot(binom.prob,
main = paste0("Binomial : Trials = ", trials, "; Probability of Success = ", p),
xlab = "Successes",
names.arg = successes,
ylab = "Probability",
ylim = c(0,1),

```

```
col = "black")
# add a horizontal line to the x axis
abline(h = 0)
```

Binomial : Trials = 10; Probability of Success = 0.5



```
# graph the cumulative binomial
cum.prob<-pbinom(0:10,10,0.5)
barplot(cum.prob,
main = paste("Cumulative; Trials = ",
              trials, "; Probability of Success = ", p),
xlab = "Successes",
names.arg = successes,
ylab = "Cumulative Probability",
ylim = c(0,1),
col = "gray")
abline(h = 0)
```

Poisson Distribution

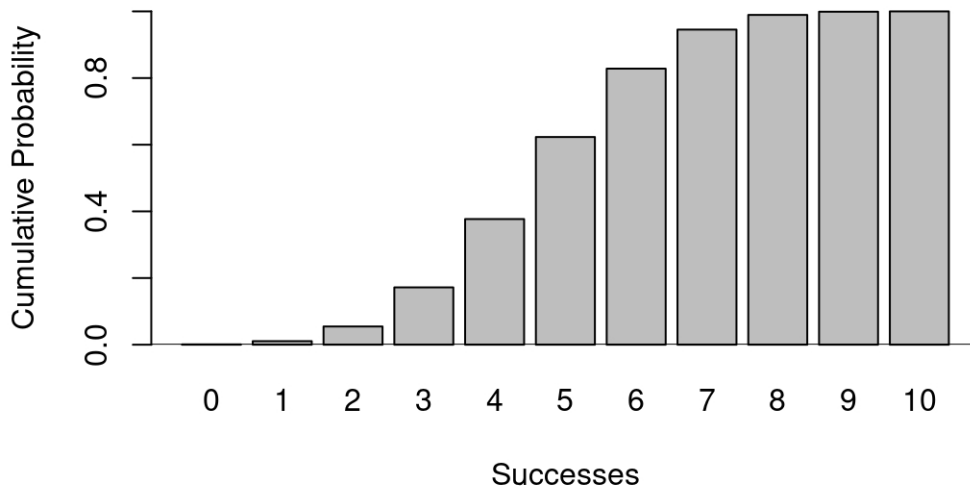
The Poisson distribution models the number of times a randomly occurring event takes place in a specified interval. For example, calls to customer service line or accidents along a busy highway or waiting period at bus stop. This distribution is specified a single parameter λ which is the average number of occurrences in a n interval. The probability mass function is given as

$$f(x) = P(X = x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x = 0, 1, 2, \dots$$

The mean and variance of the distribution both are λ . The Poisson distribution is used in many probability models and may be viewed as the limiting case of the Binomial distribution for large n with small p and $np = \lambda$.

We have There are four basic functions in *R* similar to binomial distribution.

Cumulative; Trials = 10 ; Probability of Success = 0.5



- $dpois(x, \lambda)$ returns $P(X = x)$ for Poisson distribution which is exactly x occurrence in a single interval. If x is a vector of integers, then R compute the probability for each value in x .

```
dpois(0:3,5)
[1] 0.006737947 0.033689735 0.084224337 0.140373896
```

- $ppois(x, \lambda)$ returns the probability of at most x occurrences in a single interval. In other words, it is cumulative distribution function $P(X \leq x)$.

```
ppois(0:3,5)
[1] 0.006737947 0.040427682 0.124652019 0.265025915
```

- $qpois(p, \lambda)$ returns the smallest value x such that $ppois(x, \lambda) \geq p$. That is, it computes quantiles in the specified Poisson distribution.

```
qpois(0.7,5)
[1] 6
qpois(c(0.25,0.5,0.75),5)
[1] 3 5 6
ppois(3:7,5)
[1] 0.2650259 0.4404933 0.6159607 0.7621835 0.8666283
```

- $rpois(n, \lambda)$ generates n random numbers from the this distribution with mean λ .

```
rpois(10,5)
[1] 3 6 0 4 2 8 6 3 7 5
```

Example: Calls to a customer-service line come at an average rate of 6 every 5 minutes.

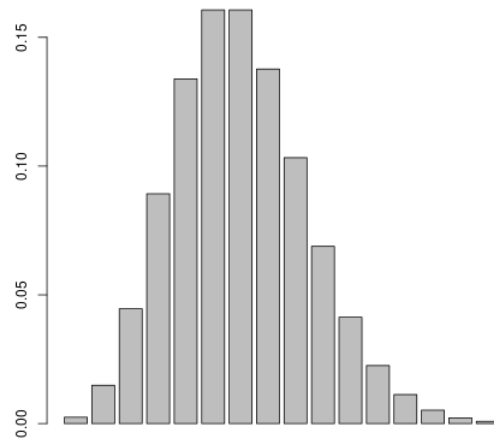
- (a) What is the probability that there are exactly 4 calls in a 5- minute period? That there are at least 4?
- (b) Construct a probability histogram for the number of calls to the line in a given 5-minute period.
- (c) Simulate the number of calls in 50 such periods.

Sol.

```
(a)      dpois(4,6)
         [1] 0.1338526
         1-ppois(3,6)
         [1] 0.8487961

(b)      x<- 0:15
         probs <- dpois(x,6)
         [1] 0.0024787522 0.0148725131 0.0446175392 0.0892350784 0.1338526175
         [6] 0.1606231410 0.1606231410 0.1376769780 0.1032577335 0.0688384890
         [11] 0.0413030934 0.0225289600 0.0112644800 0.0051989908 0.0022281389
         [16] 0.0008912556
         barplot(probs)

(c)      rpois(50,6)
         [1] 5 5 10 6 7 8 6 4 7 3 4 4 5 7 5 8 9 6 7 6
         [21] 11 10 5 6 3 8 4 7 4 9 5 5 6 2 4 9 5 4 8 6
         [41] 5 4 10 9 10 7 11 3 3 7
```



Exercises:

1. The probability of a man hitting a target is $\frac{1}{4}$. (a) If he fires 7 times then what is the probability of his hitting a target at least twice? (b) How many times must he fire so that the probability of his hitting the target at least once is greater than $\frac{2}{3}$?
2. Consider a telephone operator who, on the average, handles five calls every 3 minutes. What is the probability that there will be no calls in the next minute? At least two calls?
3. A company makes electric motors. The probability an electric motor is defective is 0.01. What is the probability that a sample of 300 electric motors will contain exactly 5 defective motors?